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Notifications 

2. Lecture 2

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 Calculator

Part A due Jan 24, 2024 20:30 IST
The following can be done after Lecture 2.

2-1

10/10 points (graded)
Each of the following equations defines an infinite collection $\mathcal{S}$ of vectors in $\mathbb{R}^2$, namely the set of vectors $\begin{pmatrix} x \\ y \end{pmatrix}$ such that x and y satisfy the equation. For which equations is this collection a vector space?

(Check all those that apply.)

☒ $x = 0$

☐ $x + y = 5$

☒ $3x + y = 0$

☐ $y = x^2$



Solution:

$x = 0$ and $3x + y = 0$.

$x = 0$: The zero vector $(0, 0)$ is in $\mathcal{S}$. Any vector in $\mathcal{S}$ has the form $(0, c)$ for some c , and multiplying by any scalar gives another vector of the same form. Adding two such vectors $(0, c)$ and $(0, d)$ gives another vector of the same form. The preceding three sentences together are what it means for $\mathcal{S}$ to be a subspace of $\mathbb{R}^2$.

$x + y = 5$: This time, $(0, 0)$ is not in $\mathcal{S}$, so the line $x + y = 5$ fails even the first condition to be a subspace.

$3x + y = 0$: The zero vector $(0, 0)$ is in $\mathcal{S}$. If (a, b) is a vector in $\mathcal{S}$ and c is a scalar, then $3a + b = 0$, so $3(ca) + (cb) = c(3a + b) = c(0) = 0$, which says that the vector $c(a, b) = (ca, cb)$ is in $\mathcal{S}$. If (a, b) and (a', b') are both in $\mathcal{S}$, then $3a + b = 0$ and $3a' + b' = 0$, so $3(a + a') + (b + b') = (3a + b) + (3a' + b') = 0 + 0 = 0$, which says that the vector $(a, b) + (a', b') = (a + a', b + b')$ is in $\mathcal{S}$. The preceding three sentences together are what it means for $\mathcal{S}$ to be a subspace of $\mathbb{R}^2$.

$y = x^2$: The zero vector $(0, 0)$ is in $\mathcal{S}$. But multiplying the vector $(1, 1) \in \mathcal{S}$ by 2 gives a vector $(2, 2)$ which is not in $\mathcal{S}$. Thus $\mathcal{S}$ fails the second condition to be a subspace. (It also fails the third condition.) Since not all three conditions were satisfied, $\mathcal{S}$ is not a subspace.

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You have used 1 of 5 attempts

Answers are displayed within the problem

2-2

10/10 points (graded)
Each of the following equations defines an infinite collection $\mathcal{S}$ of vectors in $\mathbb{R}^2$, namely the set of vectors $\begin{pmatrix} x \\ y \end{pmatrix}$ such that x and y satisfy the equation. For which equations is this collection a vector space?

Calculator

Which of the following lists is a basis for $\mathbb{R}^2$?

(Check all those that apply.)

☒ $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

☐ $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

☐ $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

☐ $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \end{pmatrix}$

☒ $\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix}$



Solution:

The bases for $\mathbb{R}^2$ are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix}$.

The list $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is a basis because both of the following hold:

- Every vector $\begin{pmatrix} a \\ b \end{pmatrix}$ in $\mathbb{R}^2$ is expressible as a linear combination of the given vectors:

$$\begin{pmatrix} a \\ b \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

- The vectors in the list are linearly independent (since neither is a linear combination of the other, i.e., neither is a multiple of the other).

$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is not a basis, because its span consists only of the scalar multiples of $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ (i.e., the line $y = x$) instead of being all of $\mathbb{R}^2$.

The list $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is not a basis, since it is not linearly independent: the third vector is a linear combination of the first two.

The list $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ is not a basis, since it is not linearly independent: the second vector is a linear combination of the first vector (which just means that it is a scalar multiple of the first vector).

The list $\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ is a basis because both of the following hold:

- Every vector in $\mathbb{R}^2$ is expressible as a linear combination of the given vectors: given any vector $\begin{pmatrix} a \\ b \end{pmatrix}$, it is possible to find x and y such that

☐ $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

☒ $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$

☒ $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$

☐ $\begin{pmatrix} 1 & 3 \\ 5 & 7 \end{pmatrix}$



Solution:

The answer is $\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$, $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, and $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$.

For a square matrix **A**, the homogeneous system **Ax = 0** has a nonzero solution if and only if the columns of **A** are linearly dependent.

For two by two matrices, the two columns are linearly dependent if one column is a scalar multiple of the other.

- The second column of $\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ is **0** times the first column.
- The second column of $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ is **−1** times the first column.
- The second column of $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ is **2** times the first column.

The matrices $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 3 \\ 5 & 7 \end{pmatrix}$ have linearly independent columns, as one is not a scalar multiple of the other.

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You have used 1 of 5 attempts

Answers are displayed within the problem

2. Lecture 2

Tonic Unit 1: Linear Algebra, Part 1 / 2 Lecture 2

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